
Topic 9: Psychological skills

Topic overview

This is a synoptic section in which students will be asked to draw on other areas of the qualification in order to understand conceptual and methodological issues. Students will develop an understanding of how to use theories and evidence from many areas of psychology and apply them to the issues.

Relevant psychological skills have been contextualised in Topics 1–8. This topic collects them together in order to ensure that all content has been covered.

Students must consider issues and debates from across all topics in order to develop a general knowledge of key issues and debates.

Subject content	What students need to learn:
9.1 Methods	9.1.1 Types of data: qualitative and quantitative data; primary and secondary data.
	9.1.2 Sampling techniques: random, stratified, volunteer and opportunity.
	9.1.3 Experimental/research designs: independent groups, repeated measures and matched pairs.
	9.1.4 Hypotheses: null, alternate, experimental; directional and non-directional.
	9.1.5 Questionnaires and interviews: open, closed (including ranked scale questions); structured, semi-structured and unstructured interviews; self-report data.
	9.1.6 Experiments: laboratory and field; independent and dependent variables.
	9.1.7 Observations: tallying; event and time sampling; covert, overt, participant, non-participant; structured observations; naturalistic observations.
	9.1.8 Additional research methods and techniques: twin and adoption studies, animal experiments, case studies as used in different areas of psychology, scanning (CAT, PET, fMRI), content analysis, correlational research, longitudinal and cross-sectional, cross-cultural and meta-analysis.
	9.1.9 Control issues: counterbalancing, order effects, experimenter effects, social desirability, demand characteristics, participant variables, situational variables, extraneous variables, confounding variables, operationalisation of variables.

Subject content	What students need to learn:
9.1 Methods <i>(continued)</i>	9.1.10 Descriptive statistics • measures of central tendency, frequency tables, graphs (bar chart, histogram, scatter diagram), normal distribution (including standard deviation), skewed distribution, sense checking data, measures of dispersion (range, standard deviation). • Produce, handle, interpret data-including drawing comparisons (e.g. between means of two sets of data). Students do not need to know formulae but are expected to be competent in simple mathematical steps.
	9.1.11 Inferential statistics Decision making and interpretation • Levels of measurement. Appropriate choice of statistical test. The criteria for and use of Mann-Whitney U, Wilcoxon, Spearman's, chi squared (for difference) tests. Directional and non-directional testing. • Use of critical value tables, one- and two-tailed testing. • Levels of significance, including knowledge of standard statistical terminology such as p equal to or greater than (e.g. $p \leq .05$). Rejecting hypotheses. Type I and type II errors. The relationship between significance levels and p values. • Observed and critical values.
	9.1.12 Methodological issues: validity (internal, predictive, ecological), reliability, generalisability, objectivity, subjectivity (researcher bias), credibility.
	9.1.13 Analysis of qualitative data (thematic analysis and grounded theory).
	9.1.14 Conventions of published psychological research: abstract, introduction, aims and hypotheses, method, results, discussion; the process of peer review.
	9.1.15 Ethical issues in research using humans (BPS Code of Ethics and Conduct, 2009), including risk assessment when carrying out research in psychology.
	9.1.16 Ethical issues in research using animals (Scientific Procedures Act 1986 and Home Office regulations).